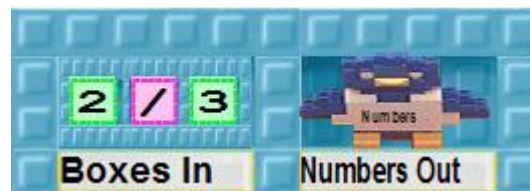


An infinite sequence of *rational numbers* between any two numbers

Pat says: There are an infinite number of *rational numbers* between any two *rational numbers*. Is Pat right?

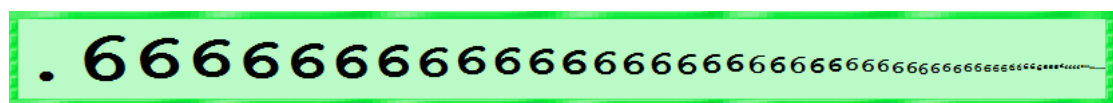
Task A: Turning boxes into numbers

Train a robot named `Box` to `Number` that takes boxes like

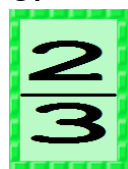


and gives a bird the number made by dividing the first number by the second. Use the pictorial instructions if you get stuck.

The result will look like



or



depending upon whether the numbers came from the Numbers notebook or Tooty.

Connect the Box to Number robot to the All Fractions team and watch the numbers given to the bird.

Are there any simplified fractions that appear in the sequence more than once?

Why or why not?

Task B: Produce the *rational numbers* from 0 to 2

Use the *Doubler* robot from Activity 1 and the robots from Activity 4. Combine them so each fraction from 0 to 1 is doubled.

Rational numbers are all numbers that can be made by dividing two

Are there any numbers greater than 2 in the sequence?
Explain...



Are there any negative numbers in the sequence?
Explain...

Joe says, "Some *rational numbers* between 0 and 2 are missing from your sequence because by doubling the numbers you are thinning them out."

Is he right?

If so, can you produce an example of a missing number?

Explain...

Suppose you had a nest that was receiving all the rational numbers between 0 and 2. It has no duplicates. If you used a robot that halved each number would the resulting sequence have any duplicates?

Explain...

Task C: Rational numbers between 0 and 1000

Train a `ThousandFold` robot that multiplies a number by 1000.

Redo Task A except use `ThousandFold` instead of `Doubler`.

What sequence do you get?

Explain...

Predict. What sequence would you get if you multiplied by 0.5 instead?

Test. Try it.

Explain.

Generalize. What sequence would you get if you multiplied by other numbers less than 1?

Generalize. What sequence would you get if you multiplied by numbers greater than 1?

Task D: *Rational numbers between 10 and 11*

Train a robot called `Add 10` that adds 10 to each incoming number before sending it out. Feed it all the rational numbers between 0 and 1.

Describe the resulting sequence.

Predict. What sequence would you get if you added -0.5 instead?

Test. Try it.

Explain.

Generalize. What sequence would you get if you replaced -0.5 with other numbers?

Task E: Generating all the *rational numbers* between any two numbers

Train a robot named *Interval* to produce all the rational numbers between *A* and *B* when given the box



If, for example, 0.6 arrives on the nest, *Interval* subtracts *A* from *B* and multiplies the result (0.5 in this example) with the incoming number (0.6 in this example). *Interval* then adds the result and *A* and gives the result (5.3 in this example) to the bird.

If you need help with *Interval* use the pictorial instructions.

Start up robots to deliver all the rational numbers between 0 and 1 to the nest of *Interval*'s box.

Will the bird in the *Rationals* hole ever be given a number less than 5?

Will it ever be given a number greater than 5.5?

Are there any numbers between 5 and 5.5 that will not be given to the bird?

Explain...

Try *Interval* with other values for *A* and *B*.

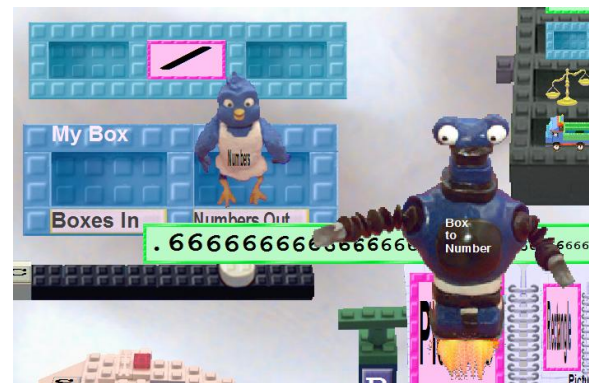
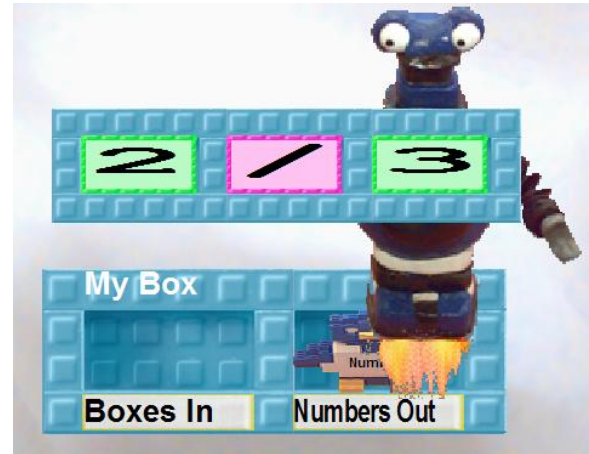
Task F: Thinking about rational numbers

Write a Web Report answering these questions:

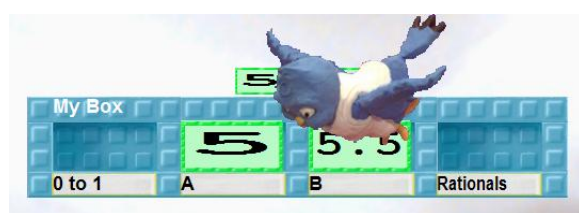
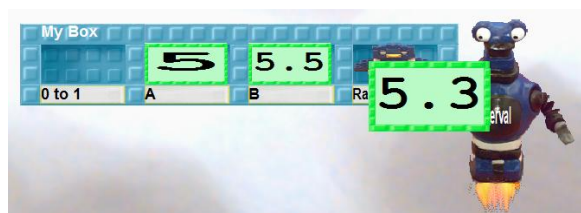
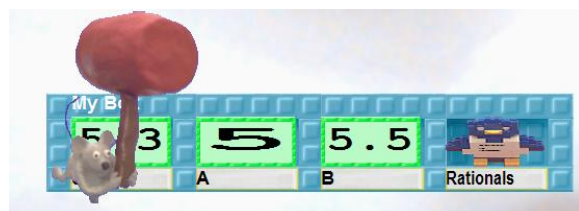
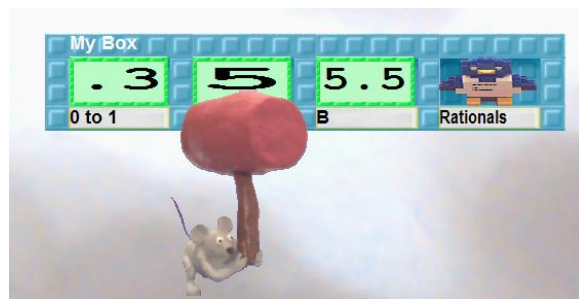
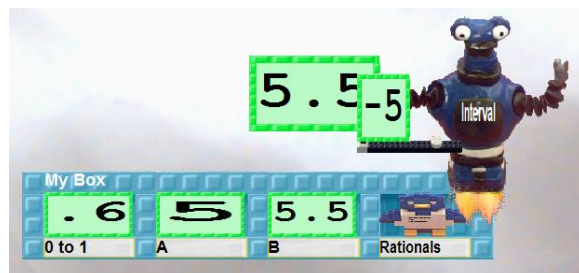
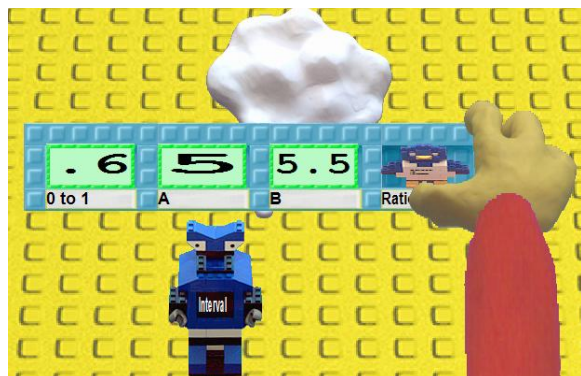
1. Is there always a number between any two *rational numbers*?
2. How many numbers are there between any two *rational numbers*?
3. Are the *rational numbers countable*?
4. Are all the rational numbers between 0 and $1/1000000$ *countable*?
5. Could all the rational numbers “dance” with all the rational numbers between 0 and $1/1000000$?

Read the other Web Reports that answer these questions. If you find reports that give answers different from your report, then add comments to their report.

Activity 7, Task A: Training Box to Number



Activity 7, Task D: Training Interval



Task F: Thinking about rational numbers